amU Aix Marseille Université

Marseille Medical Genetics

— Packaging tutorial —

Group meeting
09/03/2026

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SYSTEMS SB BIOMEDICINE

Workshop objectives

What you will see:

- Overview of development ecosystem (basics)
- Package structures and important files
- Main steps to create a package

What you will be able to do after:

- Create a basic package (python or R)
- Design “correct” structure of basic package (needed files)
- Where to find information to go deeper

Why you should create packages?

- Share codes
- Save time (reuse and maintain code between projects)
- Organize document and code
- Reproducibility
- Writing good code
- Install requisite packages automatically
- Somebody will enjoy your work!

https://py-pkgs.org/01-introduction

Development ecosystem

Package creation

- Package **structure**
- **Setup** (installation requirements ...)
- **Metadata** (license, readme ...)
- **Tests** (unit tests ...)
- **Documentation** (user manual, website ...)

Development ecosystem

Lot of helpers!

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docker

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Versioning

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Code style ...

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Development ecosystem

Package repositories

BIOCONDA

Bioconductor

python Package Index

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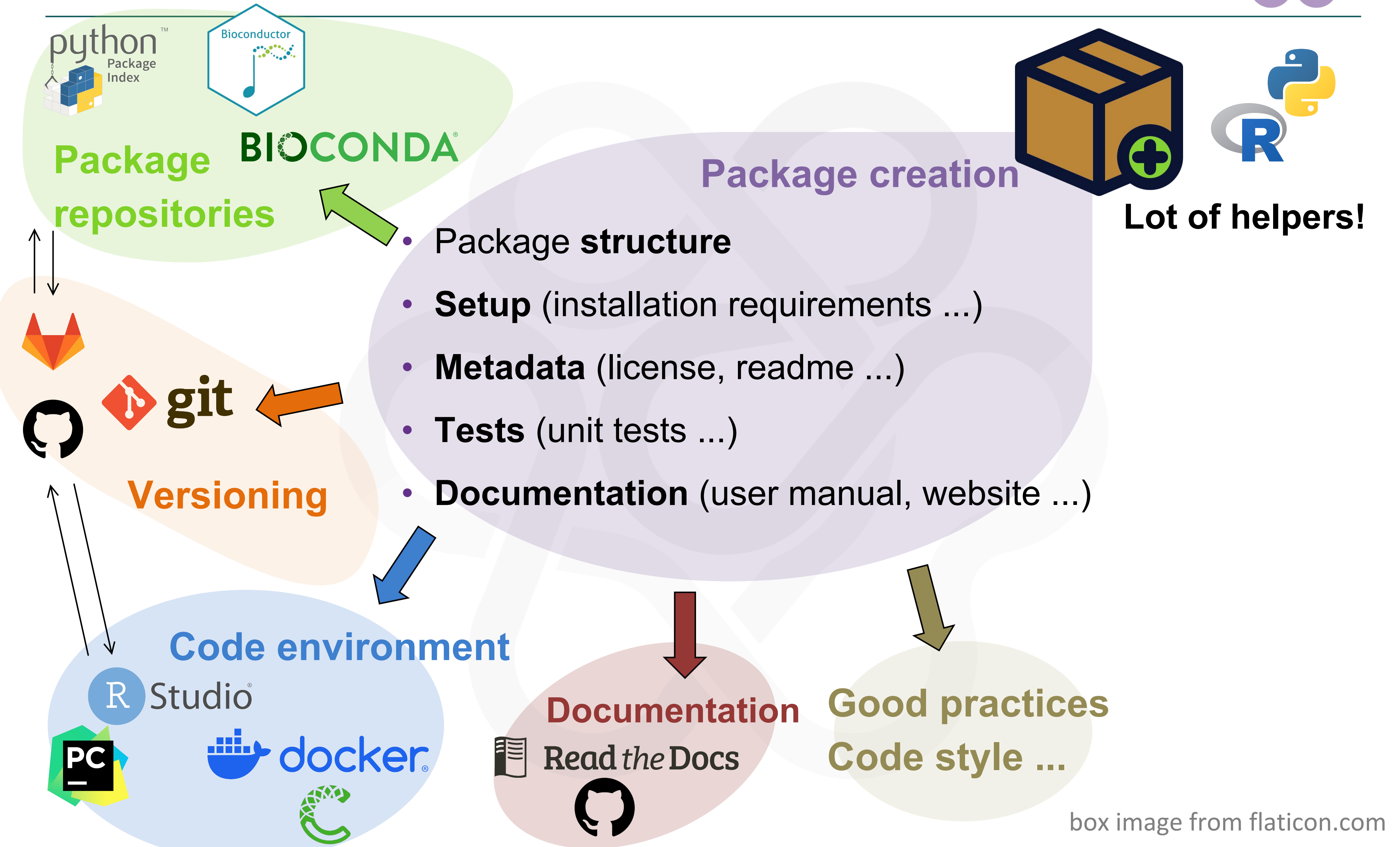

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Archiving

Software Heritage

HAL

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Code environment

Integrated Development Environment

- RStudio: https://posit.co/download/rstudio-desktop/
- PyCharm: https://www.jetbrains.com/pycharm/
- Spyder: https://www.spyder-ide.org/
- VSCode: https://code.visualstudio.com/
- Notepad++: https://notepad-plus-plus.org/

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- VSCode: https://code.visualstudio.com/
- Notepad++: https://notepad-plus-plus.org/

Package environment

- Conda: https://conda.org/
- Docker: https://www.docker.com/

Version control system

Git

- Benjamin convinced you!
- Avoid lost of work and better organisation (or not)
- Online deposits
    - GitHub: https://github.com/
    - GitLab: https://about.gitlab.com/

  ==> Make connections with others depositories

  ==> Very helpful for development

  /!\ Not a long term storage
- Github actions (Bastien RPZ)**:** https://github.com/features/actions

https://git-scm.com/

Good practices and code styles

Some standards exist

- CRAN: https://cran.r-project.org/
- Bioconductor: https://www.bioconductor.org/packages//release/bioc/vignettes/BiocCheck/inst/doc/BiocCheck.html#bioccheck-summary
- Python style:
  - PEP 8: https://peps.python.org/pep-0008/#introduction
  - PEP 257: https://peps.python.org/pep-0257/
  - Others: https://peps.python.org/

Documentation

Where to publish it and make it more findable?

- Readthedocs: https://docs.readthedocs.com/platform/stable/index.html
- Github: https://docs.github.com/fr/pages

Documentation

Where to publish it and make it more findable?

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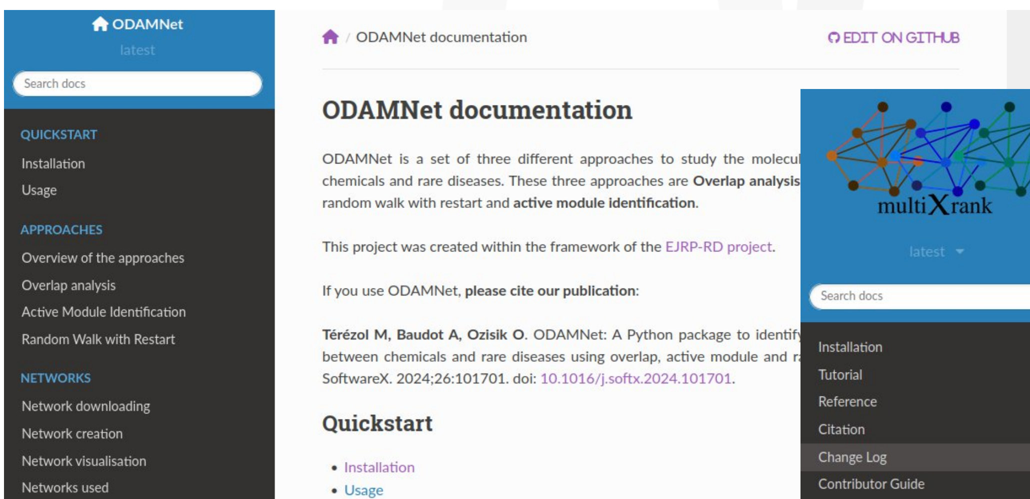
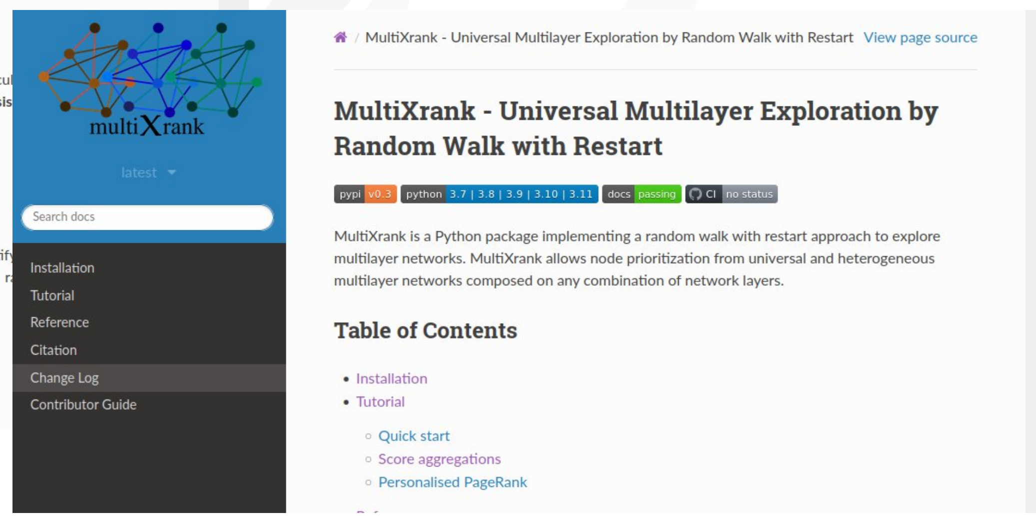
- Github: https://docs.github.com/fr/pages

- ODAMNET: https://odamnet.readthedocs.io/en/latest/
- multiXrank: https://multixrank-doc.readthedocs.io/en/latest/





Quiz!

Python-lovers question!

- How do you pronounce this?

python™
Package
Index

Quiz!

Python-lovers question!

- How do you pronounce this?

[en] *« pie pea eye »*

[fr] *« paille pie aïl »*

pie for python, pea for P (package) and eye for I (Index)
https://pypi.org/help/#pronunciation

Package repositories

Where to submit your package?

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Package repositories

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  - In R, install.package("packageName") or BiocManager::install("package")

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- **Bioconda**: https://bioconda.github.io/ (conda install –c bioconda package)

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- **Bioconda**: https://bioconda.github.io/ (conda install –c bioconda pacakge)

- Team tools:
  - orsum: https://bioconda.github.io/recipes/orsum/README.html
  - multiXrank: https://pypi.org/project/multixrank/
  - ODAMNET: https://pypi.org/project/ODAMNet/
  - MOGAMUN: https://www.bioconductor.org/packages/release/bioc/html/MOGAMUN.html

Code archiving

Software Heritage

- Collect, preserve and share software
- Safe place for long term preservation
- **SWHID** (Permanent identifier)
- GitHub/GitLab/CRAN/Pypi ...

https://www.softwareheritage.org/

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- Submit to HAL

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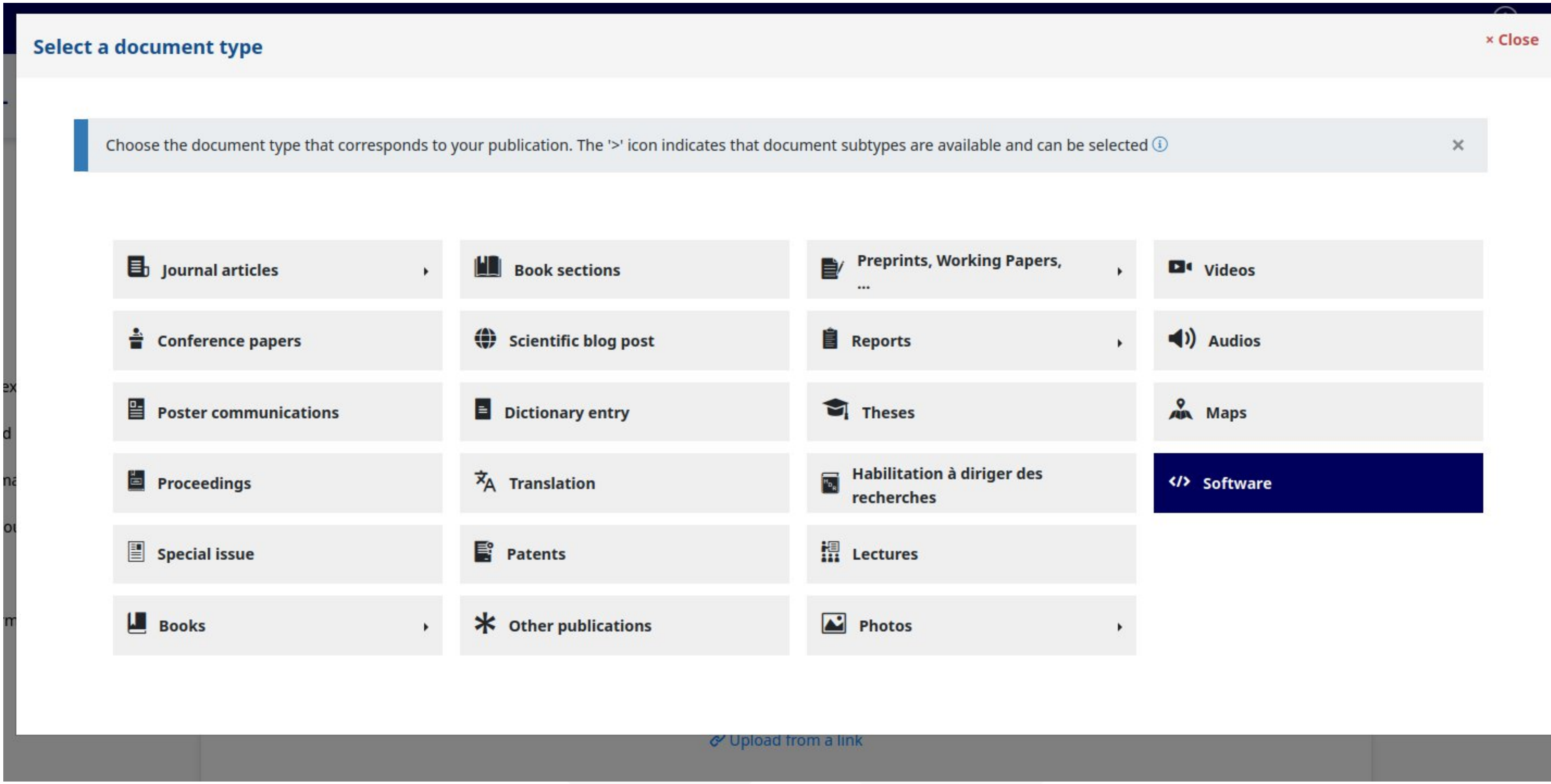
https://hal.science/

Code archiving

HAL submitting page



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Repository structure of the project

Main components

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  README
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pyproject.toml

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inspired from
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COOKIECUTTER

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METADATA files

Describe package and how to use it

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Journée prodev:
https://www.prodev.cnrs.fr/prodev-2025-neuvieme-rencontre-du-reseau-des-developpeurs-de-provence/

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Useful for installation

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- **pyproject.toml**: equivalent to DESCRIPTION

SETUP files

Useful for installation – project.toml file of MOFA2 python

```
[build-system]
requires = ["poetry-core"]
build-backend = "poetry.core.masonry.api"

[tool.poetry]
name = "mofapy2"
version = "0.7.3"
description = "Multi-omics factor analysis"
authors = ['Ricard Argelaguet', 'Damien Arnol', 'Danila Bredikhin', 'Britta Velten']
license = "LGPL-3.0"
readme = 'README.md'
repository = "https://github.com/bioFAM/mofapy2"
homepage = "https:/biofam.github.io/MOFA2/"

[tool.poetry.dependencies]
python = "^3.8"
numpy = "*"
pandas = "*"
scipy = "~1"
scikit-learn = "~1"
h5py = "~3"
anndata = { version = "> 0.8", optional = true }

[tool.poetry.dev-dependencies]
pytest = "^7.4.2"
anndata = "~0.9"
```

SETUP files

Useful for installation – DESCRIPTION file of MOFA2 R

```
Package: MOFA2
Type: Package
Title: Multi-Omics Factor Analysis v2
Version: 1.21.3
Maintainer: Ricard Argelaguet <ricard.argelaguet@gmail.com>
Authors@R: c(person("Ricard", "Argelaguet", role = c("aut", "cre"),
                     email = "ricard.argelaguet@gmail.com",
                     comment = c(ORCID = "http://orcid.org/0000-0003-3199-3722")),
              person("Damien", "Arnol", role = "aut",
                     email = "damien.arnol@gmail.com",
                     comment = c(ORCID = "http://orcid.org/0000-0003-2462-534X")),
              person("Danila", "Bredikhin", role = "aut",
                     email = "danila.bredikhin@embl.de",
                     comment = c(ORCID = "https://orcid.org/0000-0001-8089-6983")),
              person("Britta", "Velten", role = "aut",
                    email = "britta.velten@gmail.com",
                    comment = c(ORCID = "http://orcid.org/0000-0002-8397-3515"))
              )
Date: 2023-01-12
License: file LICENSE
Description: The MOFA2 package contains a collection of tools for training and analysing multi-omic factor analysis (MOFA).
Encoding: UTF-8
Depends: R (>= 4.0)
Imports: rhdf5, dplyr, tidyr, reshape2, pheatmap, ggplot2, methods, RColorBrewer, cowplot, ggrepel, reticulate, HDF5Array, g
Suggests: knitr, testthat, Seurat, SeuratObject, ggpubr, foreach, psych, MultiAssayExperiment, SummarizedExperiment, SingleC
biocViews: DimensionReduction, Bayesian, Visualization
URL: https://biofam.github.io/MOFA2/index.html
BugReports: https://github.com/bioFAM/MOFA2
VignetteBuilder: knitr
LazyData: false
StagedInstall: no
NeedsCompilation: yes
RoxygenNote: 7.3.3
SystemRequirements: Python (>=3), numpy, pandas, h5py, scipy, argparse, sklearn, mofapy2
```

WRITE CODE
Functions and comments

- Repository for coding part (src/ or R/)

  => Better to use a repository

WRITE CODE

Functions and comments

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- Make **functions** (one per file, groups per file ...)
- **Comment** functions (title, parameters, description, examples ...)

WRITE CODE

Functions and comments

- Repository for coding part (src/ or R/)

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- Make **functions** (one per file, groups per file ...)
- **Comment** functions (title, parameters, description, examples ...)
- Code style:
  - Existing standards (PEP 8 ...)
  - Styler package in R
  - Recommendation from Bioconductor
- Specificities of languages

DOCUMENT

Explain what package is doing and how

- **Automatic generation** of the documentation
- Function documentation
- Function examples
- README
- User manual

  => Build automatic docs

  => Build automatic website

TESTS

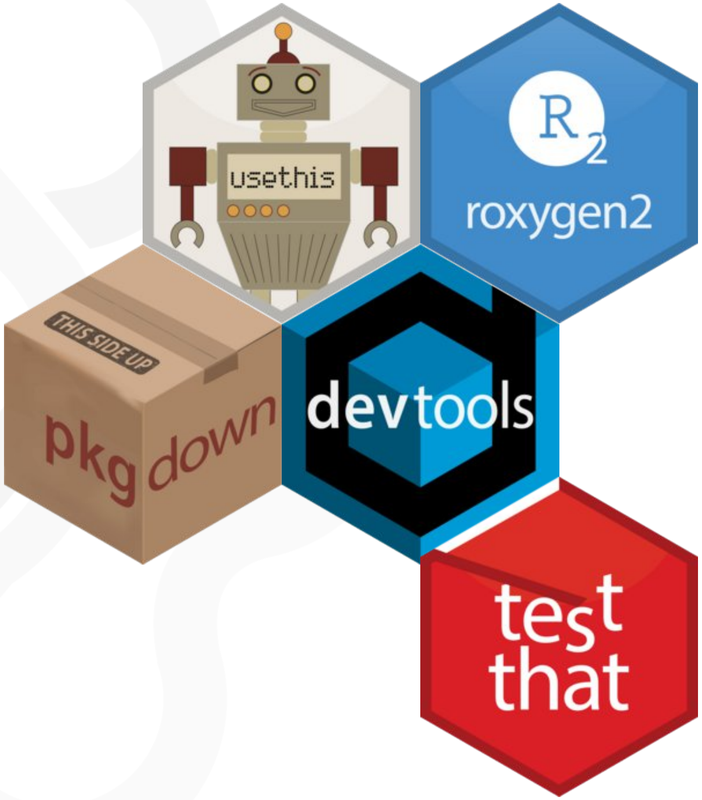
Is package working well?

- Different type of test (e.g. unit tests)
- Test every function (is better)
- For unit test, comparison between what the function is doing and what the result expected
- Recommanded (expected for R)

R – package development needed

Non exhaustive list

- Automate package and project setup:
  - devtools
  - usethis
- Unit tests:
  - testthat
- Automate documentation:
  - roxygen2
- Website:
  - pkgdown



Python – package development needed

Non exhaustive list

- Automate package and project setup:
  - cookiecutter
  - poetry / setuptools
- Unit tests:
  - pytest
- Automate documentation:
  - sphinx
- Website:
  - sphinx

Useful links

- R packages - 2edition: https://r-pkgs.org/ (**the Bible of R packages**)

- Bioinfo.net – Blog post in french : https://bioinfo-fr.net/pourquoi-et-comment-deposer-un-package-r-sur-bioconductor
- Bioconductor: https://contributions.bioconductor.org/
- Paula Moraga : https://www.paulamoraga.com/blog/2022-04-12-rpackages.html

Python
Useful links

- For R-lovers: https://py-pkgs.org/
- For the package philosophy: https://docs.python-guide.org/ (The Hitchhiker's guide to Python)
- For updated resources:
  - Python Packaging User Guide: https://packaging.python.org/en/latest/
  - Packaging Python Projects: https://packaging.python.org/en/latest/tutorials/packaging-projects/

- Charles Tapley Hoyt:
  - https://cthoyt.com/teaching/
  - https://www.youtube.com/watch?v=lo_g-GbYtaA
  - https://www.youtube.com/watch?v=f6brWkO9OiE (I learnt with this one)
  - https://cthoyt.com/2020/06/03/how-to-code-with-me-organization.html
  - https://cthoyt.com/2020/04/25/how-to-code-with-me-flake8.html

Your turn now!

You will create a package in your favourite language!

Workshop session

Step one

1. Choose a language: R or Python
2. Install all the required packages
3. Create a folder where you want to work in your computer
4. Open you favourite IDE
5. Now, **create a package!**

Workshop session

Step two

- Package name: **hello**
- Short description: **Display the current date and time**
- Version: **0.1.0**
- Authors: **your name**
- License: **MIT**
- 3 functions:
  - currentTime()
  - displayTime(current_time, style)
  - TT()

References

Team tools

- **orsum** - Ozisik O, Térézol M, Baudot A. orsum: a Python package for filtering and comparing enrichment analyses using a simple principle. BMC Bioinformatics. 2022 Jul 23;23(1):293. doi: 10.1186/s12859-022-04828-2.
- **ODAMNET** - Térézol M, Baudot A, Ozisik O. ODAMNet: A Python package to identify molecular relationships between chemicals and rare diseases using overlap, active module and random walk approaches. SoftwareX. 2024;26:101701. doi: 10.1016/j.softx.2024.101701.
- **MOGAMUN** - Novoa-del-Toro EM, Mezura-Montes E, Vignes M, Térézol M, Magdinier F, Tichit L, et al. (2021) A multi-objective genetic algorithm to find active modules in multiplex biological networks. PLoS Comput Biol 17(8): e1009263. https://doi.org/10.1371/journal.pcbi.1009263
- **multiXrank** - Baptista, A., González, A., Baudot, A. Universal multilayer network exploration by random walk with restart. Commun Phys 5, 170 (2022)

References

Tools

- **MOFA2** - Argelaguet R, Velten B, Arnol D, Dietrich S, Zenz T, Marioni JC, Buettner F, Huber W, Stegle O (2018). “Multi-Omics Factor Analysis—a framework for unsupervised integration of multi-omics data sets.” Molecular Systems Biology, 14. doi:10.15252/msb.20178124.

Others useful links

- **Python environment management**: https://www.linkedin.com/feed/update/urn:li:activity:7432314684441047040/?updateEntityUrn=urn%3Ali%3Afs_updateV2%3A%28urn%3Ali%3Aactivity%3A7432314684441047040%2CFEED_DETAIL%2CEMPTY%2CDEFAULT%2Cfalse%29
- **Python style code**: https://www.linkedin.com/posts/basilemarchand_pythontooling-activity-7434857420771733504-_eZG?utm_source=share&utm_medium=member_desktop&rcm=ACoAABbWcs8Bon7eTnW7skjFB-UEDpZy7Ol6_fI